

1 Fundamental Quantities & the Mole

- Avogadro's number $N_A = 6.022 \times 10^{23}$ particles per mole.
- 1 mole = N_A particles = molar mass expressed in grams.
- 1 amu = $\frac{1}{12}$ (mass of a ^{12}C atom) = 1.66×10^{-24} g.
- Gram-atomic mass = mass of 1 mol atoms; gram-molecular mass = mass of 1 mol molecules.

2 The Mole — Master Relations

The central bridge between mass, number, and volume of a substance:

$$n = \frac{w}{M} = \frac{N}{N_A} = \frac{V_{\text{gas at STP}}}{22.4 \text{ L}} = \frac{PV}{RT}$$

- Number of particles = $n \times N_A$; mass $w = n \times M$ (g).
- M = molar mass (g/mol). For a gas at STP, 1 mol occupies 22.4 L (273 K, 1 atm).

3 Average Atomic Mass & Isotopes

- Average atomic mass $\bar{A} = \frac{a_1A_1 + a_2A_2 + \dots}{100} = \frac{\sum a_iA_i}{100}$, a_i = % abundance of isotope i .
- Average molar mass of a mixture = $\frac{\text{total mass}}{\text{total moles}} = \sum x_iM_i$, x_i = mole fraction.

4 Percentage Composition

- % of an element = $\frac{(\text{no. of atoms of that element}) \times (\text{atomic mass})}{\text{molecular mass of compound}} \times 100$.
- Mass of that element in w g of compound = $w \times \frac{\% \text{ element}}{100}$.

5 Empirical & Molecular Formula

- Empirical formula:** simplest whole-number ratio of atoms.
- Steps: % $\rightarrow$ divide by atomic mass $\rightarrow$ divide by smallest $\rightarrow$ whole-number ratio.
- Molecular formula** = $n \times (\text{empirical formula})$, where $n = \frac{\text{molar mass}}{\text{empirical formula mass}}$.

6 Stoichiometry & Limiting Reagent

- A balanced equation gives the **mole ratio** of reactants and products.
- Limiting reagent:** the reactant that yields the least product; it is consumed completely.
- Identify it by dividing moles of each reactant by its coefficient — the smallest value limits.
- POAC** (conservation of atoms): moles of any element are conserved across the reaction.

7 Concentration Terms

TERM	DEFINITION
Molarity M	$\frac{\text{moles of solute}}{\text{volume of solution (L)}} = \frac{w_{\text{solute}} \times 1000}{M_{\text{solute}} \times V_{\text{mL}}}$
Molality m	$\frac{\text{moles of solute}}{\text{mass of solvent (kg)}} = \frac{w_{\text{solute}} \times 1000}{M_{\text{solute}} \times w_{\text{solvent(g)}}$
Mole fraction x	$x_A = \frac{n_A}{n_A + n_B}, \quad x_A + x_B = 1$
Normality N	$\frac{\text{gram equivalents}}{\text{volume (L)}} = (n\text{-factor}) \times M$
Percent w/w	$\frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \times 100$
Percent w/v	$\frac{\text{mass of solute (g)}}{\text{volume of solution (mL)}} \times 100$
ppm	$\frac{\text{mass of solute (g)}}{\text{mass of solution (g)}} \times 10^6$
Strength (g/L)	$\frac{\text{mass of solute (g)}}{\text{volume of solution (L)}} = M \times M_{\text{solute}}$

Molality, mole fraction and %w/w are **temperature-independent**; molarity and %w/v change with temperature.

NOTE: ppm can also be expressed in terms of volume.

8 Interconversion of Concentrations

- Molarity from density d (g/mL) and mass %:
$$M = \frac{10 d (\%w/w)}{M_{\text{solute}}}$$
- Molarity $\rightarrow$ molality:
$$m = \frac{1000 M}{1000 d - M M_{\text{solute}}}$$
- Mole fraction $\rightarrow$ molality:
$$m = \frac{x_{\text{solute}} \times 1000}{x_{\text{solvent}} \times M_{\text{solvent}}}$$
- Strength (g/L) = $M \times M_{\text{solute}} = N \times E$.

9 Dilution & Mixing

- Dilution (same solute): $M_1 V_1 = M_2 V_2$ (equivalently $N_1 V_1 = N_2 V_2$).
- Mixing two solutions of the same solute:
$$M_{\text{mix}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$$
- On dilution the moles of solute stay constant; only the volume changes.

10 Equivalent Concept & n -factor

- Equivalent weight $E = \frac{\text{molar mass } M}{n\text{-factor}}$; gram-equivalents = $\frac{w}{E} = n \times (n\text{-factor})$.

SPECIES	<i>n</i> -FACTOR
Acid	basicity (no. of replaceable H^+)
Base	acidity (no. of replaceable OH^-)
Salt	total charge on cation (= total charge on anion)
Oxidising / reducing agent	electrons gained / lost per formula unit

11 Normality & Law of Equivalence

$$N = (n\text{-factor}) \times M = \frac{\text{gram equivalents}}{V(\text{L})}$$

- Law of equivalence:** in any reaction the equivalents of each species are equal.

$$\text{eq of A} = \text{eq of B} \implies N_1 V_1 = N_2 V_2 \text{ (at the end point)}$$

$$\text{For } aA + bB \rightarrow \text{products: } \frac{n_A}{a} = \frac{n_B}{b} \text{ (mole basis)} \equiv \text{eq}_A = \text{eq}_B$$

12 *n*-factor & Redox Half-Reactions

n-factor = change in oxidation number (electrons transferred) per formula unit.

Permanganate KMnO_4

- Acidic: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ ($n = 5$)
- Neutral / weakly basic: $\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{MnO}_2 + 4\text{OH}^-$ ($n = 3$)
- Strongly basic: $\text{MnO}_4^- + \text{e}^- \rightarrow \text{MnO}_4^{2-}$ ($n = 1$)

Dichromate $\text{K}_2\text{Cr}_2\text{O}_7$ (acidic)



Common reducing agents

- Oxalate: $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2 + 2\text{e}^-$ ($n = 2$)
- Ferrous: $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ ($n = 1$)
- Peroxide (as reductant): $\text{H}_2\text{O}_2 \rightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ ($n = 2$)
- Thiosulphate: $2\text{S}_2\text{O}_3^{2-} \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{e}^-$ ($n = 1$ per $\text{S}_2\text{O}_3^{2-}$)

13 Gas Stoichiometry & Ideal Gas

- Ideal gas: $PV = nRT \Rightarrow n = \frac{PV}{RT}$.
- Molar volume = **22.4 L** at STP (273 K, 1 atm); **22.7 L** at 273 K, 1 bar (newer IUPAC).
- Gas density $\rho = \frac{PM}{RT}$.
- Vapour density $\text{VD} = \frac{M_{\text{gas}}}{2} \Rightarrow M_{\text{gas}} = 2 \times \text{VD}$.
- Avogadro's law: at the same T, P , equal volumes contain equal moles ($V \propto n$).

$$R = 0.0821 \text{ L}\cdot\text{atm}\cdot\text{K}^{-1}\cdot\text{mol}^{-1} = 8.314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1} = 2 \text{ cal}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$$

14 Eudiometry (Gas Analysis)

Gas volumes react in the ratio of their moles (Avogadro's law), so stoichiometric coefficients = volume ratios.

General hydrocarbon combustion


- On cooling, H_2O condenses (its volume ignored); CO_2 is absorbed by KOH.
- Volume contraction = (initial gaseous volume) – (final gaseous volume).

15 Key Reactions for Mole-Concept Problems
Decomposition (source of O_2 / CO_2)

- $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2 \uparrow$
- $CaCO_3 \xrightarrow{\Delta} CaO + CO_2 \uparrow$
- $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2 \uparrow$

Metal + acid (source of H_2)

- $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2 \uparrow$
- $2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2 \uparrow$
- $Mg + 2HCl \longrightarrow MgCl_2 + H_2 \uparrow$

Precipitation (gravimetry)

- $AgNO_3 + NaCl \longrightarrow AgCl \downarrow + NaNO_3$
- $BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 \downarrow + 2HCl$

16 Special Quantitative Analyses
Volume strength of H_2O_2

- $2H_2O_2 \longrightarrow 2H_2O + O_2 \uparrow$; " V volumes" = 1 mL gives V mL O_2 at STP.
- Normality = $\frac{V}{5.6}$, Molarity = $\frac{V}{11.2}$, % strength (w/v) = $\frac{17V}{56}$.

Oleum (fuming H_2SO_4)

- $SO_3 + H_2O \longrightarrow H_2SO_4$; labelled $> 100\%$ = mass of H_2SO_4 formed on adding water to 100 g oleum.
- % free $SO_3 = \frac{40(P - 100)}{9}$, where P = labelled %.

Hardness of water

- Hardness (ppm) = $\frac{\text{mass of } CaCO_3 \text{ equivalent}}{\text{mass of water}} \times 10^6$.

17 Quick Recall / Common Data

- $N_A = 6.022 \times 10^{23}$; molar volume = 22.4 L at STP (22.7 L at 1 bar).
- 1 amu = 1.66×10^{-24} g; $n = \frac{w}{M} = \frac{N}{N_A} = \frac{V_{STP}}{22.4} = \frac{PV}{RT}$.
- $N = (n\text{-factor}) \times M$; $E = \frac{M}{n\text{-factor}}$; $N_1V_1 = N_2V_2$.
- $M = \frac{10d(\%w/w)}{M_{\text{solute}}}$; $M_{\text{gas}} = 2 \times VD$; Strength = $M \times M_{\text{solute}}$.
- Temperature-independent: molality, mole fraction, %w/w.